AAKRUTI Innovation Competition

AAKRUTI 2026 · INDIA GEO FINALE · ROBOTICS

AquaSentinel

A river that raises its own alarm.

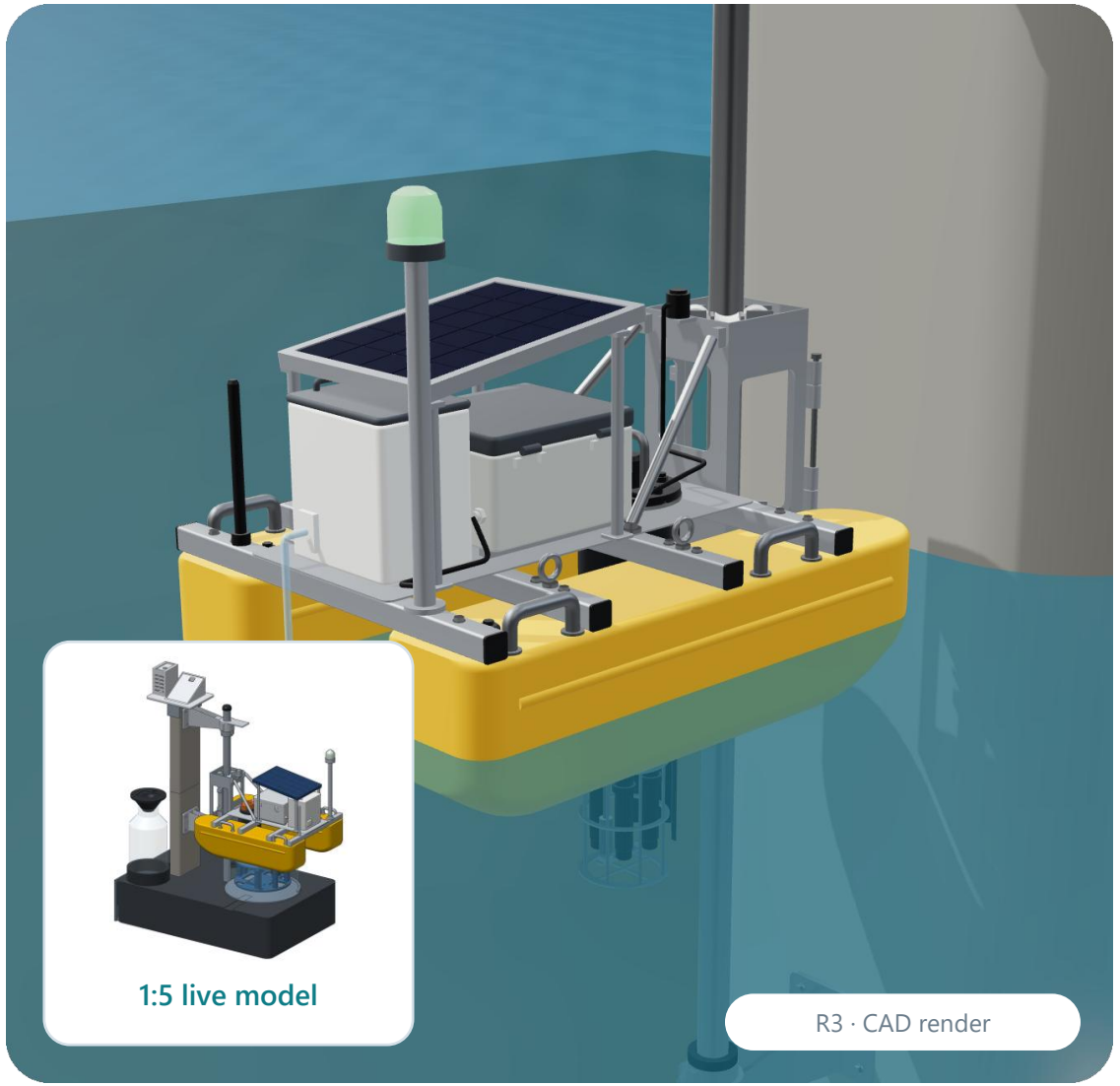
A floating sentinel on a bridge pier. Every 15 minutes it reads the water; the moment a spill passes, it turns its beacon red, sends an SMS and bottles a sample for the lab.

Senses every 15 min

Decides on board

Acts alarm + sample

Team Sanjivani Hero's Journey · AG26-1500$41
Sanjivani University, Kopargaon, Maharashtra



THE HIDDEN CRISIS

Every night, someone counts on nobody watching.



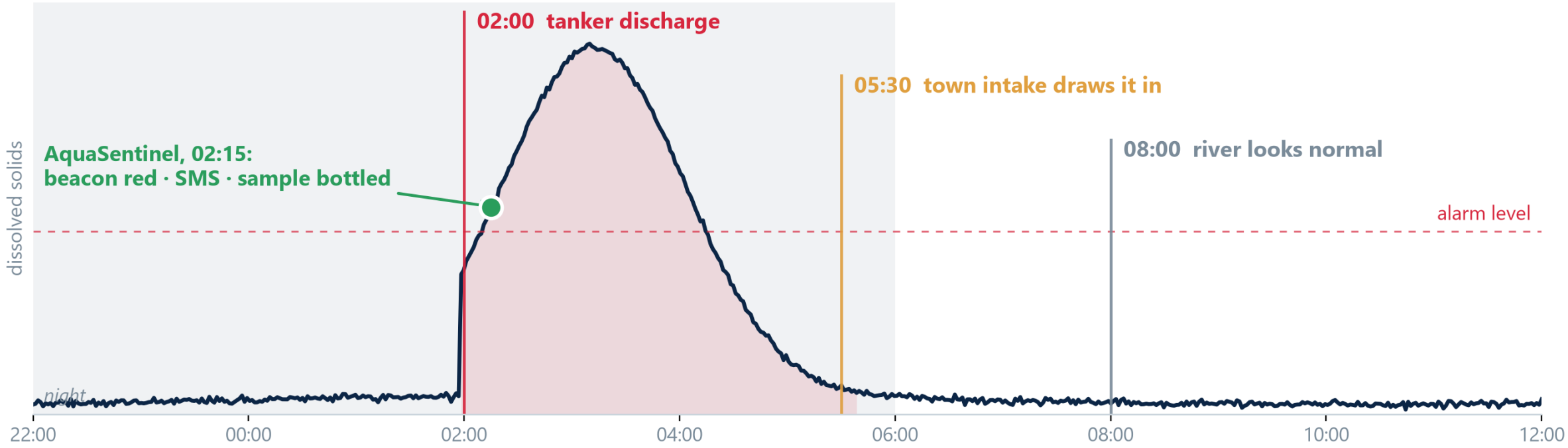
Untreated waste reaches the river at night, between samples, when no one is looking.

Scene from our journey video (illustrative)

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THE PROBLEM

2:00 a.m. A spill passes. By sunrise, the river looks clean.



Families downstream drink from the town intake

Farmers irrigate with the same water

River life takes the hit first

Illustration of a typical night-time discharge, not measured data.

WHY IT'S MISSED

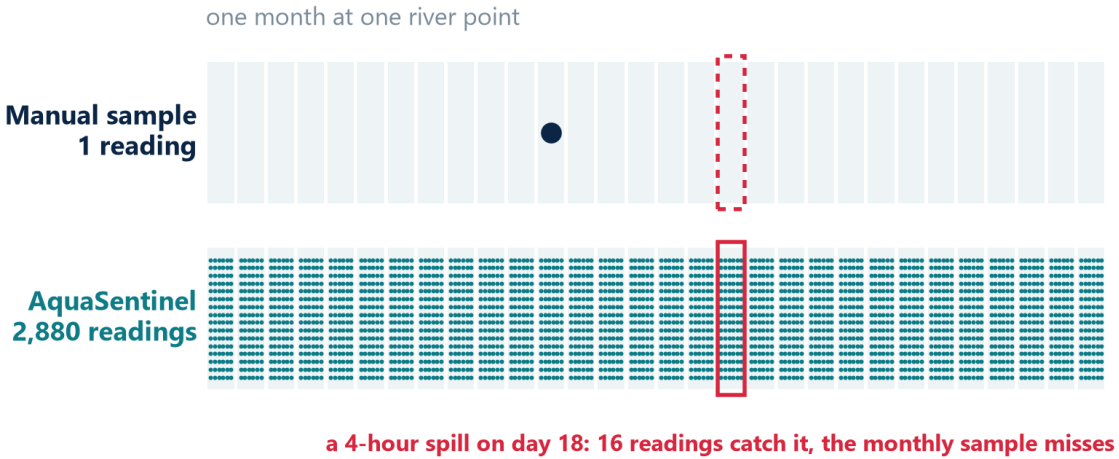
Rivers are checked once a month. Spills last hours.

296
polluted river stretches and locations in India

54
of them in Maharashtra: the most of any state

1 / month
at best: manual grab samples at most monitoring points

₹35–40 L
for one real-time station, so there are very few



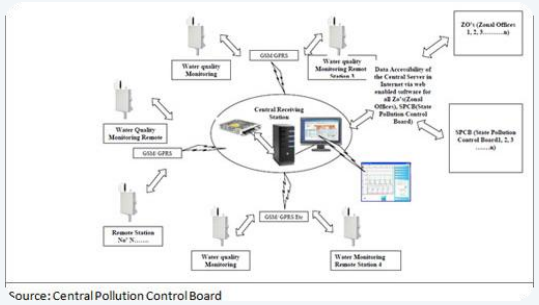
A spill that lasts a few hours falls between two samples.

In a month, a manual sample gives 1 reading. AquaSentinel gives 2,880.

Sources: CPCB assessment of polluted river stretches, Sept 2025 (2022–23 data); CPCB reply to NGT, OA 413/2025; CSE (station cost).

WHAT EXISTS TODAY

Good tools exist. None of them does the whole job.



Source: Central Pollution Control Board

CPCB real-time station

✓ 17 parameters, every 15 min

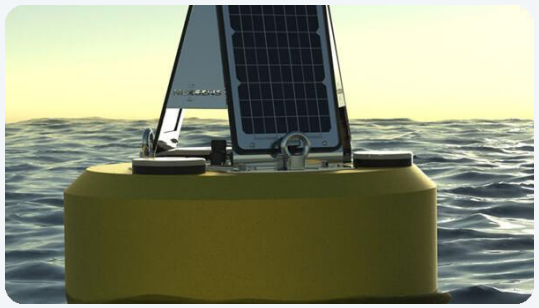
✗ ₹35–40 lakh each, so very few sites



YSI EXO2 sonde

✓ reference-grade sensors

✗ needs a mount, logger and telemetry



NexSens CB-450 buoy

✓ solar buoy that carries sondes

✗ anchored; no beacon, no sampler



Teledyne ISCO 6712 sampler

✓ bottles samples on a trigger

✗ no continuous readings of its own

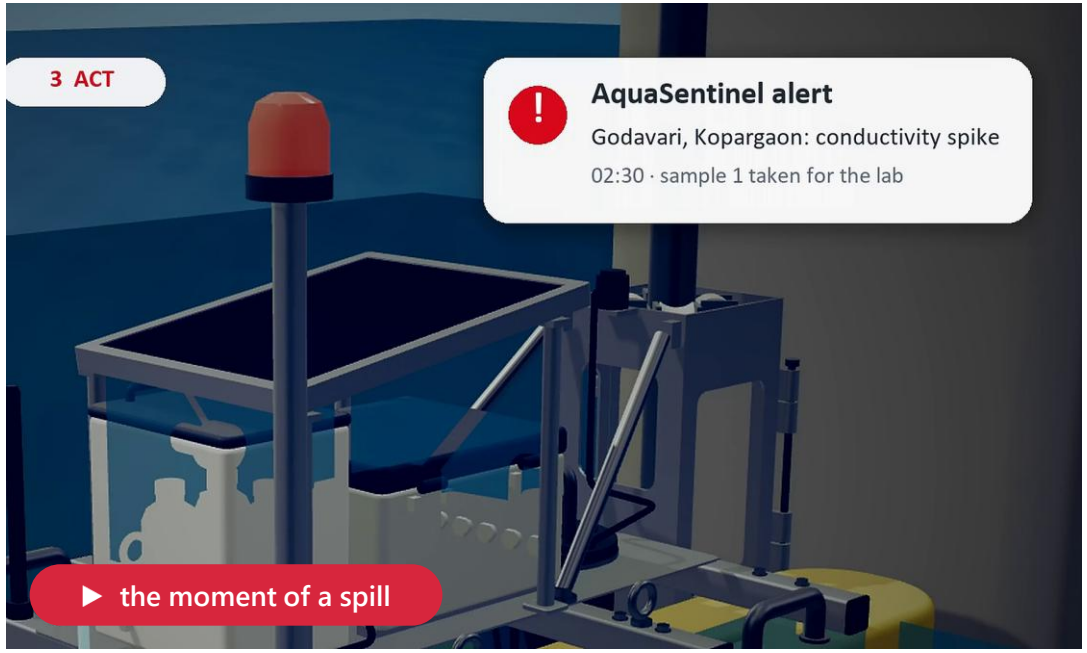
AquaSentinel does all four on one float: reads every 15 min · decides on site · warns on the bank · bottles the spill · ≈ ₹48k each

Product facts from manufacturer and distributor pages (Oct 2026); CSE for the station cost. Full comparison in the backup slides.

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OUR INNOVATION

A sentinel that lives in the river and decides on the spot.



Green while the water is normal. Red in one cycle when a spill passes, and the SMS goes out.

SENSE every 15 minutes

pH, dissolved solids, turbidity and water temperature; rain, light and humidity; vapour from the sample bottle

DECIDE on board, no cloud

A Raspberry Pi compares each reading with hard limits and with what is normal for that hour of the day

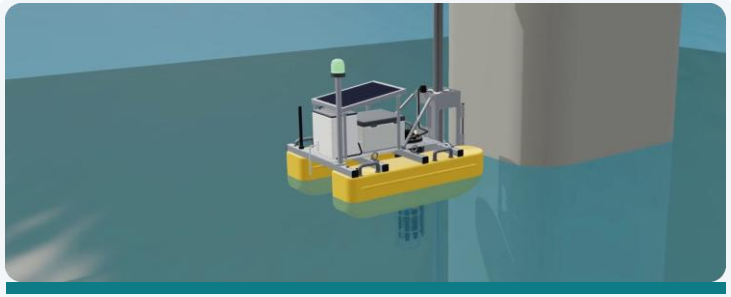
ACT in seconds

A red beacon for people on the bank, an SMS to the water-works, and a bottle of the spill for the lab

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WHAT'S NEW

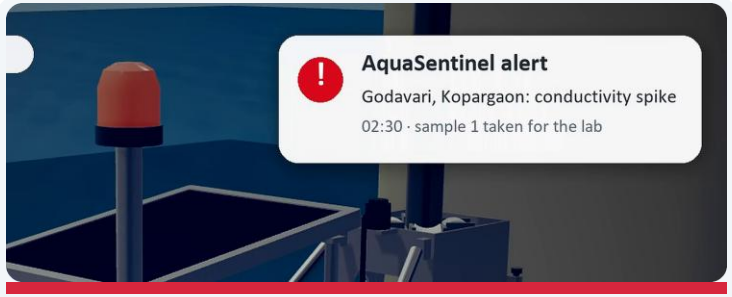
Three ideas, working together in one low-cost float.



1 Rides the river

A float on a guide pole clamped to a bridge pier. The river lifts it 2 m: no anchors, no motor, no riverbed work.

→ **installed and serviced from the bridge**



2 Names the spill

The pattern across its sensors says industrial, acidic, runoff or solvent. A broken sensor shows FAULT, never "normal".

→ **the officer knows whom to call**



3 Keeps the evidence

It bottles the water at the moment of the spill, so the lab tests the right water, not next month's.

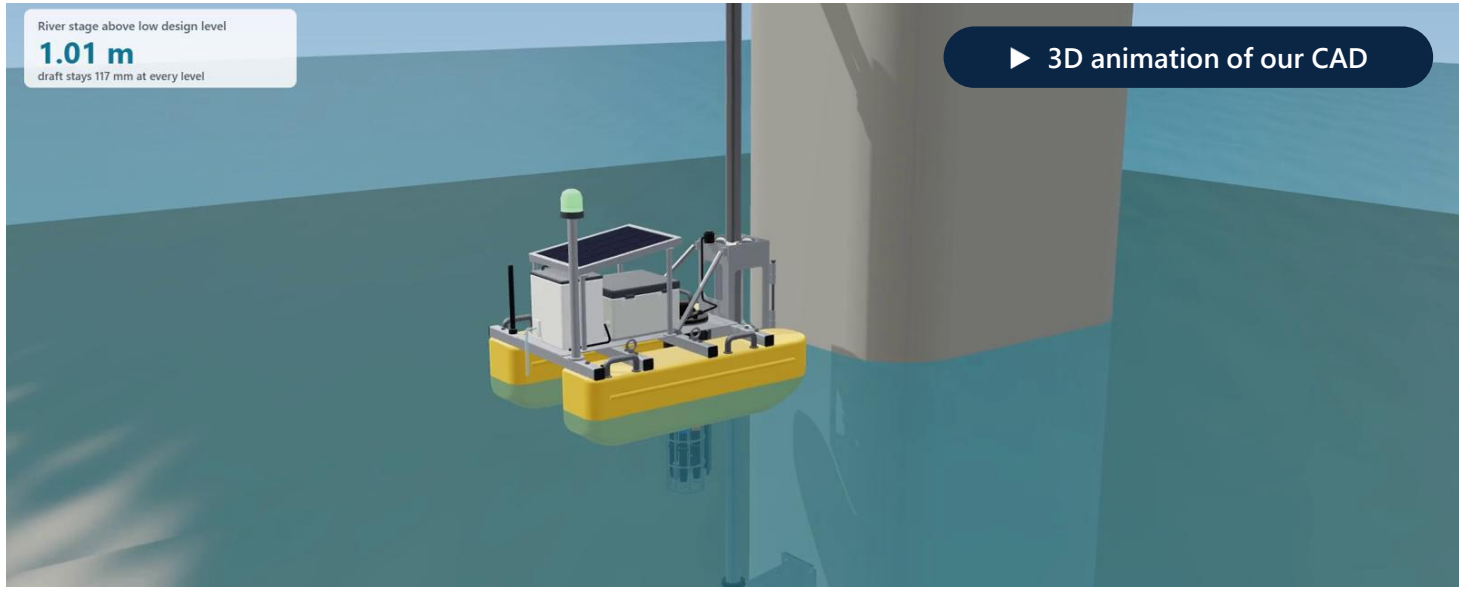
→ **a lead for the pollution board**

≈ 75 sentinels for the price of one real-time station ≈ ₹48,000 each in production vs ₹35–40 lakh (a station measures more)

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INNOVATION 1 · IT RIDES THE RIVER

No anchors, no motor: the river lifts it.



River rises 2 m and falls back · the current turns the float · the carriage gate opens for service

2 m of river rise

0 drive motors

8 rollers on the pole

3D animation and design checks on the full-size R3 CAD (66 parts). Static load case; impact and anchors still to verify.

14 / 14 design checks pass on the R3 CAD model

2.66 pole safety factor: the most conservative case we found

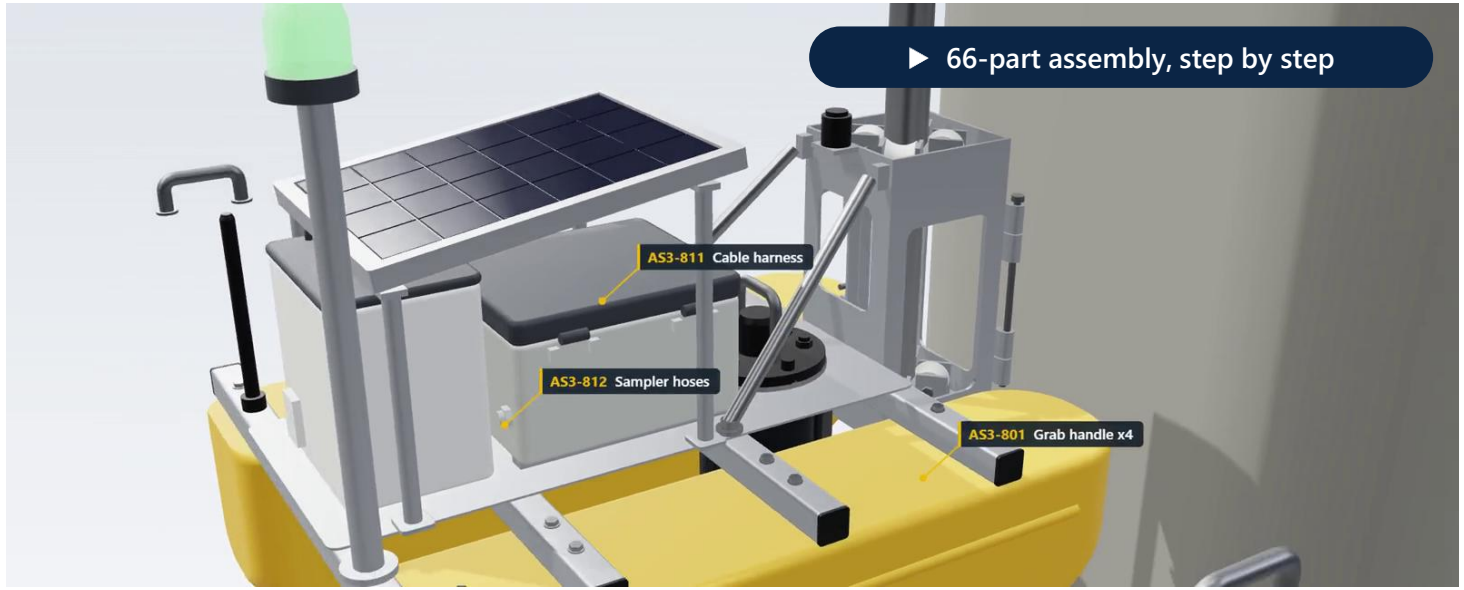
143 mm freeboard; 2.27× reserve buoyancy

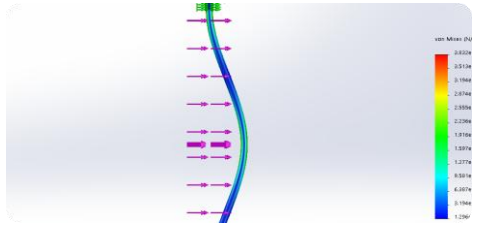
1.5 m/s flood current + 500 N debris load, designed for

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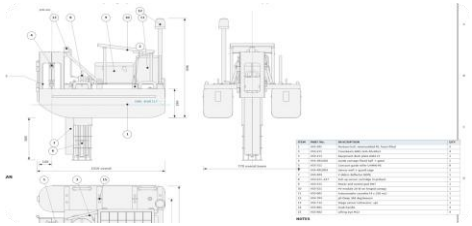
CAD AND SIMULATION

Assembled and simulated in SOLIDWORKS.

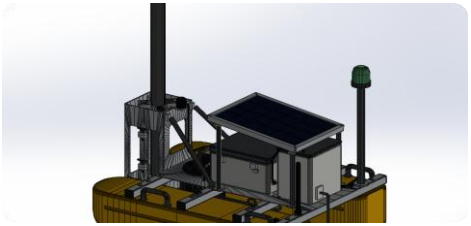




Simulation: pole stress (FEA)



Drawing set with BOM



The SOLIDWORKS assembly

66 parts — one assembly, a material and part number on every part

14 checks — buoyancy, stability and clearances, computed on the CAD

2 studies — SOLIDWORKS Simulation of the pole and the pier bracket

6 sheets — drawing set, from general arrangement to details

17 new parts — designed in SOLIDWORKS for the 1:5 live model

Pack and Go — the full package, plus a STEP file for every part

Also on 3DEXPERIENCE: 3DSwym · ENOVIA Project Planner

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INNOVATION 2 · IT NAMES THE SPILL

It doesn't just measure. It recognises what happened.

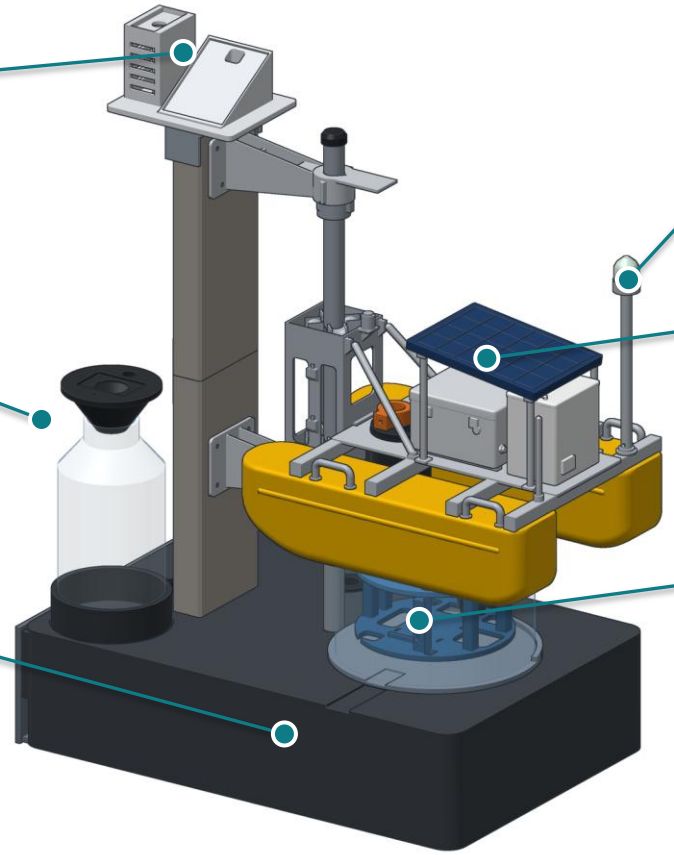
	pH	Dissolved solids	Turbidity	Vapour	Rain	AquaSentinel says
Industrial discharge	·	▲▲	·	·	·	INDUSTRIAL
Acid spill	▼▼	·	·	·	·	ACIDIC
Farm runoff after rain	·	·	▲	·	●	RUNOFF
Solvent or fuel	·	·	·	▲	·	SOLVENT
A sensor fails	?	·	·	·	·	SENSOR FAULT

How it decides: hard limits (pH < 6 or > 9, dissolved solids > 1,400 ppm) and a baseline it learns for every hour; 2 unusual readings out of 3 make an event.

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PROOF · OUR 1:5 LIVE MODEL

We printed it at 1:5, around the real sensors.



Rendered from our print files; the 17 new parts are native SOLIDWORKS parts.

37 printed parts, 17 designed new for this model

1:1 real sensors, not mock-ups

4 water samples you can swap in seconds

✓ designed | ✓ printed | wiring + code | live 19 Nov

status at submission, 7 Nov 2026

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THE ELECTRONICS

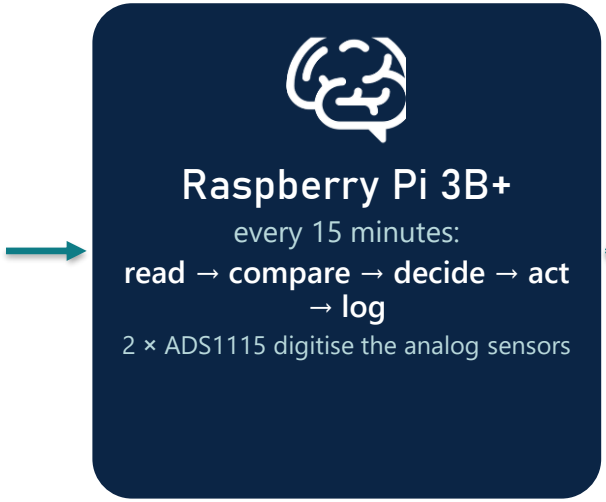
One Raspberry Pi. Eight sensors. Three actions.

IN THE WATER

- **pH** — acid or alkaline
- **Dissolved solids** — salts, effluent
- **Turbidity** — mud, sediment
- **Water temp.** — hot discharge

ABOVE THE WATER

- **Rain** — is it raining?
- **Light** — day or night
- **Air temp. + humidity** — weather
- **MQ-3 vapour** — solvent or fuel



- **Beacon turns red** — for anyone on the bank
- **SMS alert** — water-works + board (SIM800L)
- **Sample bottled** — pump fills 250 mL for the lab

WHY WE MOVED FROM ESP32 TO RASPBERRY PI

	ESP32 (semi-final design)	Raspberry Pi 3B+ (now)
Hardware	industrial RS485 probes still to buy	parts our team already has, on the bench
Software	C++ firmware: logic only	Python: logic, data log and SMS on one board
Room to grow	limited memory	camera for floating plastic, 4G modem
Electronics cost	≈ ₹80,000	**≈ ₹11,000 in production**

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SEE IT LIVE · OUR BOOTH, 19 NOV

Four waters, one honest answer each.



Clean tap water

readings stay steady

1

NORMAL



Salty water

dissolved solids jump

2

INDUSTRIAL



Vinegar

pH falls

3

ACIDIC



Muddy water

turbidity rises, salts flat

4

RUNOFF



Sanitiser in the bottle

vapour rises

5

SOLVENT



We unplug a probe

a reading is lost

6

SENSOR FAULT

A 0.5 L dish is not a river: the model proves the whole chain, from sensor to SMS. It classifies a change; the bottle goes to the lab to name the chemical.

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WHO IT SERVES, AND HOW IT PAYS

Built for the people who must act first.

Water-works operator

closes the intake before the spill reaches town

Pollution board officer

knows when and where to sample

Farmers and irrigation

a record of the water they use

Industrial estate

independent proof of a clean discharge

HOW THE MONEY WORKS (ONE SENTINEL)

₹48,000

to build one sentinel

production cost at 100 units, ex-GST (first prototype ≈ ₹65,000)

₹8,000 / month

paid by the board or water-works

covers the sentinel, installation, monthly service, alerts and 4 lab samples a year

≈ 12 months

to pay back the hardware

then about 40 % margin over a 5-year contract

Kopargaon pilot · 3 floats → MPCB regional programme → Maharashtra's 54 polluted sites → 1 station = 75 sentinels

Cost: value-engineered bill of materials, 7 Nov 2026 (estimate). Fee and roll-out are proposals to validate in the pilot.

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OUR JOURNEY SO FAR

What we have built, and where it goes next.

✓ Problem researched
CPCB data, 296 sites, the Godavari at Kopargaon

✓ 3 design generations
v1 → v2 → R3; two big mistakes caught in CAD

✓ 66-part R3 in SOLIDWORKS
a material and part number on every part

✓ 14 / 14 design checks
plus 2 SOLIDWORKS Simulation studies

✓ Drawing set + Pack and Go
6 sheets, a STEP file for every part

✓ Decision logic proven on data
4 of 4 spills flagged in a 30-day test

✓ 1:5 live model printed
37 parts, 17 designed new, real sensors

✓ Electronics designed
Raspberry Pi, 8 sensors, pin map, power budget

✓ Costed for production
≈ ₹48k a sentinel; service model

19–20 Nov
live at our booth: four waters, six checks

Next
full-size tank test: 2 m travel, rollers

With permits
30-day Godavari pilot at Kopargaon

Feb 2027
AAKRUTI international finale, if selected

Status at submission, 7 Nov 2026. Electronics wiring and code for the live model are under way for the booth.

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Watching the river every 15 minutes

1. Earlier warning for people downstream
2. A sample that tells the lab where to look
3. A light anyone on the bank can read



Soham Kadu
Project Lead



Prasad Ugale
System Design



Bhakti Kadam
Electronics Lead



Tanmay Pendbhaje
Software Lead



Dr. Kiran Wakchaure
Mentor

Thank you. We'd love your questions.

Sanjivani Hero's Journey · Sanjivani University, Kopargaon · AG26-1500$41

Questions? Click a topic

A1 All 14 design checks pass

A2 Pole safety factor: we quote the lowest

A3 Power: about 3 days without sun

A4 Decision logic on the Raspberry Pi

A5 Cost: ≈ ₹48,000 a sentinel in production

A6 Market: a large problem, sparse monitoring

A7 Risks we know, and how we handle them

A8 Roadmap to the finale and the pilot

A9 Prototype levels: the 1:5 LIVE model

A10 Materials and manufacturing

Full comparison: existing solutions vs AquaSentinel

	CPCB real-time station	YSI EXO2 sonde	NexSens CB-450 buoy	ISCO 6712 sampler	AquaSentinel (R3 design + 1:5 LIVE model)
Readings, parameters	every 15 min; ≈ 17 incl. BOD, COD	user-set; up to 6 sensors	set by its logger; carries sondes	samples only; reads a sonde via SDI-12	every 15 min; pH, TDS, turbidity, water + air temp., rain, light, sample vapour
Mounting	fixed station; layout n.p.v.	user-supplied mount, pipe or buoy	moored, 3 eye-nuts	bank or manhole, intake hose	float on a guide pole clamped to a bridge pier
River-level change	n.p.v.	depends on mount	floats within mooring scope	hose reach (max lift per spec)	rides 2 m on 8 rollers, no drive motor (calculated)
Decision on site	data to central server	logs; alarm logic not listed	logger and web portal; alarms n.p.v.	event trigger on up to 16 inputs	on board (Raspberry Pi): limits + learned baseline + simple classes
Local light / remote alert	n.p.v.	not part of the sonde	n.p.v.	modem dial-out alarm option	red/green beacon + SMS (2G)
Physical sample	n.p.v.	no (sensor)	not in listed product	yes, up to 24 bottles	250 mL per event (designed; 1 bottle on the model)
Indicative cost	₹35–40 lakh instruments (CSE)	price on request	not listed	not listed	≈ ₹48k in production, ₹65k prototype (ex-GST)
Power, upkeep	calibration, batteries (CSE)	4 D-cells, ≈ 90 days at 15 min	3 × 15 W solar	12 V DC battery or mains	20 W solar + LiFePO4, ≈ 3 days dark (estimate); monthly visit

What we combine differently (narrow, to be proven): each capability exists separately, and a buoy + sonde + triggered sampler could be integrated. Our difference is the architecture: a pier-guided float that follows river level without a mooring, an on-board decision, a beacon for people on the bank and one bottle per event, built from low-cost parts at about ₹48,000. Not validated in a river yet.

Sources as on slide 5: CPCB · CSE · YSI EXO2 · NexSens CB-450 · ISCO 6712 · n.p.v. = not publicly verified. AquaSentinel values are design estimates (backup A1–A5).

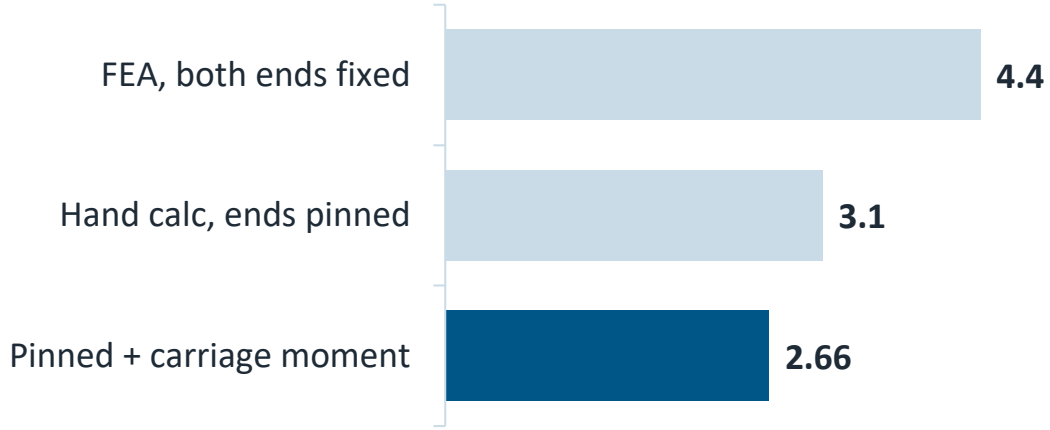
A1 · All 14 design checks pass

Check	Result	Criterion
Reserve buoyancy	2.27×	≥ 2.0×
Freeboard	143 mm	≥ 100 mm
Transverse GM	0.55 m	≥ 0.15 m (IMO reference)
Righting arm	positive to 60°, max 180 mm at 25°	positive to 60°
Pole safety factor, pinned	3.1 (2.66 with carriage moment)	≥ 2.0
Pole deflection, pinned	5.8 mm	≤ span / 200 (13.4 mm)
Jamming, rollers (μ 0.05)	margin 8.6	no self-locking
Jamming, seized rollers (μ 0.25)	margin 1.7	no self-locking
Float lag at full design load	17.7 mm	≤ 30 mm
Weathervane stability	lateral centre 351 mm aft of the pole	aft of the pole
Pole corridor clearance	9.7 mm	≥ 5 mm
Float vs fixed parts, 2 m travel	no clash	none
Only rub rails touch the pier	rub rail 12 mm outboard	rub rails outermost
Part-to-part interference	none in 101 pairs	none

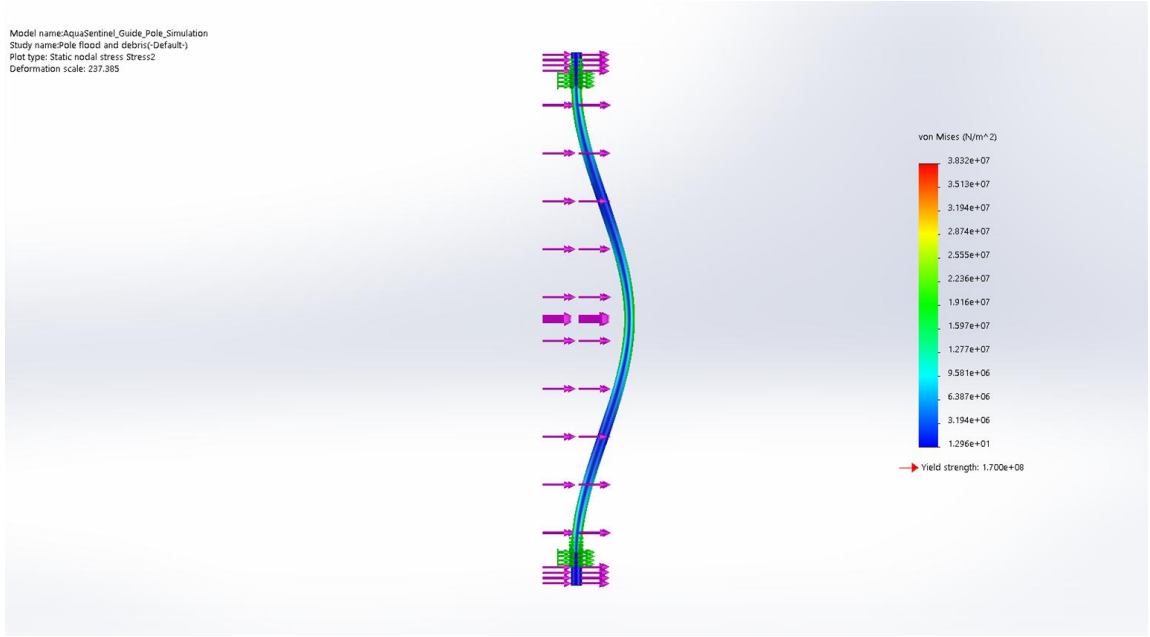
Source: 05_R3_FULL_SIZE_CAD/Design_Checks, computed from the R3 CAD; masses partly catalogue estimates.

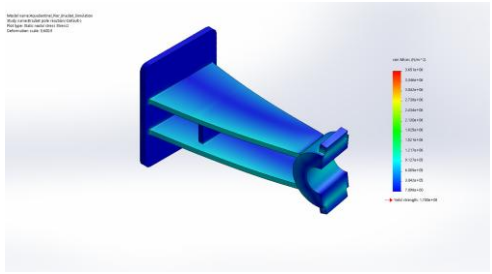
A2 · Pole safety factor: we quote the lowest

Pole safety factor by idealisation (requirement ≥ 2.0)



- Peak stress in MPa: FEA 38.3 · pinned 54.8 · pinned + moment 64.0 (316L yield 170)
- Load: 642 N at the float = drag ½·1000·1.5²·1.05·0.120 m² = 142 N + 500 N debris (quasi-static), plus drag on the pole
- Not yet included: debris impact factor, fatigue, vortex shedding

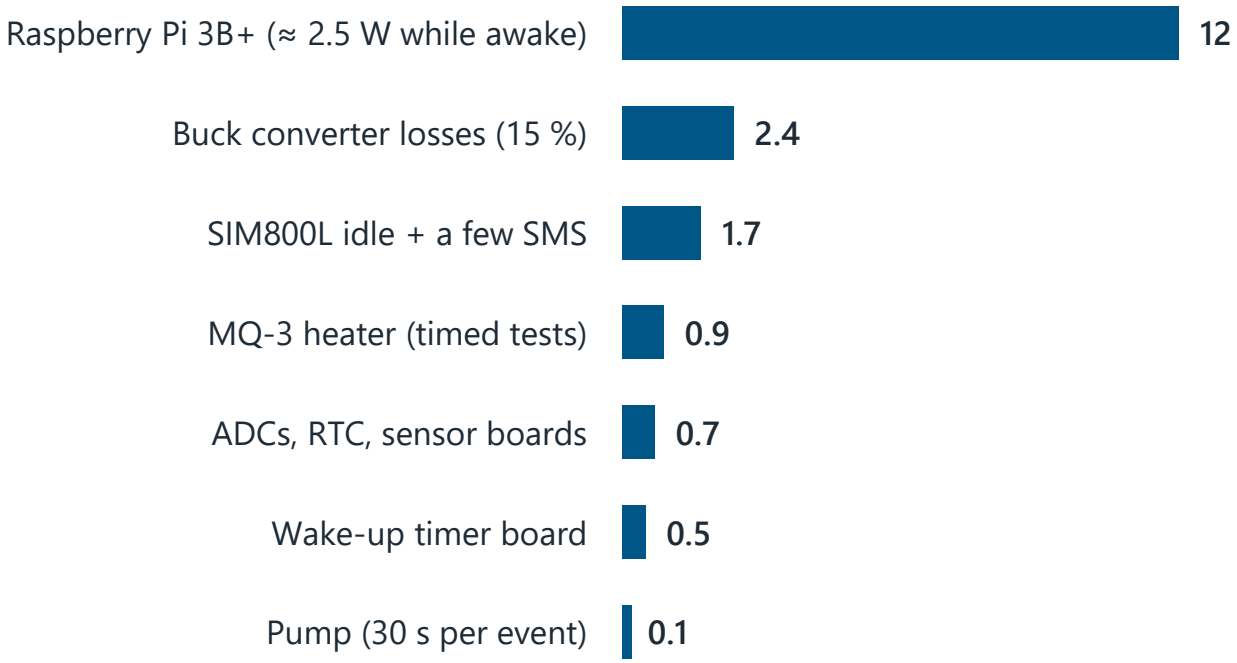




Bracket: 3.7 MPa peak under the 430 N reaction. Anchors, welds and debris impact still to check.

A3 · Power: about 3 days without sun (estimate)

Daily energy, Pi woken 3 min every 15 min, Wh per day (design estimate)



Source: 03_ELECTRONICS_RASPBERRY_PI design, section 5.3. Nothing measured yet; at the booth the model runs on mains.

≈ 19 Wh/day
timed mode: the Pi sleeps between 15-min readings

≈ 100 Wh/day
if the Pi stays on: needs 50 W solar + 20 Ah

61 Wh
usable: 12.8 V 6 Ah LiFePO4 at 80 % depth of discharge

≈ 3 days
with no sun in timed mode (61 ÷ 19)

39 Wh/day
20 W × 3.0 sun-hours on a monsoon day, after 35 % losses

A4 · Decision logic on the Raspberry Pi

How it decides (designed)

- Every 15 minutes: read all sensors first, then act, so the pump and the GSM radio never disturb the analog readings
- Hard limits act at once: pH < 6 or > 9; TDS > 1,400 ppm (≈ EC 2,250 µS/cm, CPCB Class E); turbidity 3× its baseline
- A baseline for every hour of the day; an EVENT when 2 of the last 3 readings are unusual
- Simple classes from the pattern: TDS↑ = saline/industrial · pH↓ = acidic · turbidity↑ with TDS flat = runoff (rain plate wet: stronger) · MQ-3↑ in the bottle = solvent vapour
- First 7 days: baseline learning only. An early-warning system, not a chemical identifier: the lab confirms

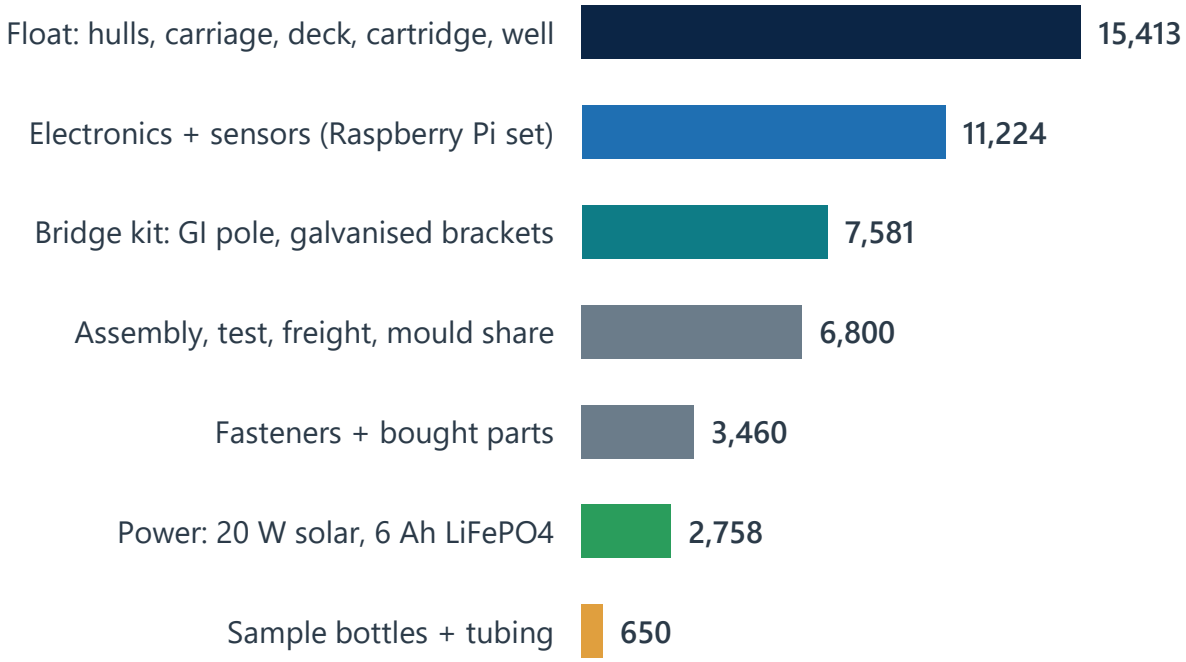
Rules we build in from our audit

- A failed or impossible reading sets SENSOR FAULT, never "normal water"
- pH and TDS boards are never powered together (they disturb each other); read one, then the other
- The learned baseline is saved to disk, so a reboot doesn't erase it
- One bottle per event; the event clears after 4 clean readings
- Status: being coded in Python for the 19 Nov demo

Sources: 03_ELECTRONICS_RASPBERRY_PI design (sections 7–8); our audit of the semi-final decision engine (18 Oct).

A5 · Cost: ≈ ₹48,000 a sentinel in production

One sentinel, float + bridge kit, ₹ ex-GST, at 100 units (estimate, 7 Nov 2026)



Total ≈ ₹47,900 (≈ ₹56,500 with GST) · first prototype ≈ ₹64,600 · before cost-down ₹90,800

Source: 07_COST_AND_BUSINESS/03_Value_Engineered_Cost_48k. Rates: GI ₹70–85/kg + galvanising ₹16–35/kg (Indian listings). Estimates: get 3 quotes.

−₹42,900 (−47 %)
Bridge parts 316L stainless → hot-dip galvanised steel; welded aluminium → bolted aluminium + HDPE; wiper and cool-box dropped

Same float mass
steel replaces stainless of the same density; HDPE parts are lighter, so buoyancy and stability checks still hold

Stronger pole
GI pipe yields at ≥ 210 MPa vs 170 MPa for 316L: the quoted safety factor 2.66 stays conservative

Trade-off
annual coating check on galvanised parts; CAD still shows 316L until the production model is updated

A6 · Market: a large problem, sparse monitoring

Fact	Number	Source
Polluted river stretches/locations, India	296 on 271 rivers: 156 stretches + 140 locations (CPCB, published 2025, 2022–23 data)	PIB, 16 Mar 2026; CPCB
Maharashtra	54, still the most of any state (55 in the 2022 assessment)	PIB; CPCB
National monitoring network	≈ 4,922 locations, 2,265 on rivers; manual samples, monthly or quarterly	CPCB reply to NGT, OA 413/2025
Real-time station cost	₹35–40 lakh for 10-parameter instruments; ≈ ₹50–84 lakh with systems	CSE; Lok Sabha reply 5115

Illustrative market size (assumptions)

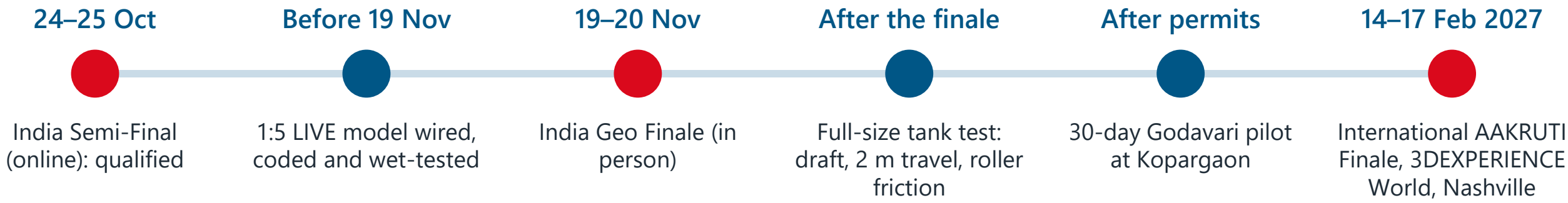
Scope	Floats (3 per counted site)	Revenue at ₹0.96 lakh per float-year (₹8,000 a month)
Maharashtra, 54 sites	162	≈ ₹1.56 crore a year
India, 296 sites	888	≈ ₹8.5 crore a year

Market size is illustrative: floats per stretch and the fee (₹8,000 a month) are assumptions to validate in the pilot.

A7 · Risks we know, and how we handle them

Risk	Why it matters	How we handle it
Flood above 2 m of travel	the float is held under at the top stop	get gauge data (CWC/WRD); longer travel or a breakaway link; lift out on flood warnings
Galvanised steel in water	coating wears; rust on bridge parts	hot-dip galvanising, stainless fasteners, annual coating check
Sensor accuracy and drift	hobby-grade probes drift; false or missed alarms	alerts on sudden change, not absolute values; pH 4/7 and TDS standard at every monthly visit; industrial probes as an option
2G network switch-off	SIM800L would lose SMS	drop-in 4G module (≈ ₹1,900); the beacon still works locally
Sample is not legal evidence	Water Act §21 needs the Board's procedure	a triage sample that triggers the Board's formal sampling
pH and TDS in the same water	powered probes disturb each other	switch each board's supply; read one, then the other
Permits	no pilot without them	bridge owner, river authority and MPCB: start now

A8 · Roadmap to the finale and the pilot



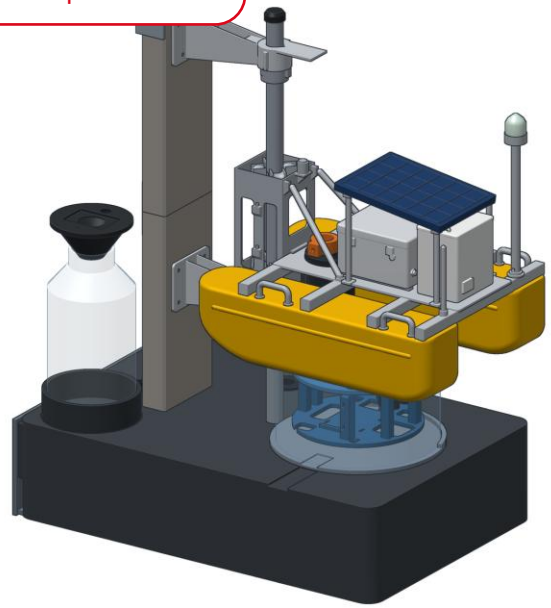
● competition dates (organisers) ● our plan

What we aim to show at the 19–20 Nov finale (our plan)

- The 1:5 LIVE model at our booth: real sensors in water, the Raspberry Pi deciding every cycle
- Four water samples (clean, salty, acidic, muddy) and a solvent test in the sample bottle
- A probe unplugged: SENSOR FAULT, never green

A9 · Prototype levels: the 1:5 LIVE model

Render, not a photo
made from our print files



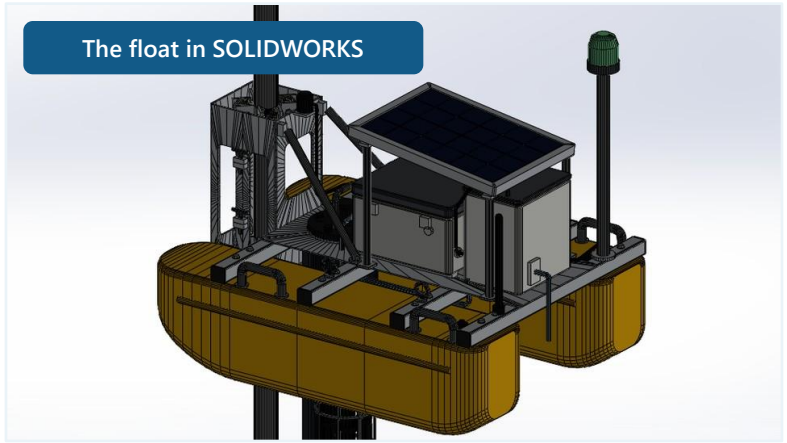
1:5 LIVE model: R3 float on its pole, real sensors under it in a dish, Raspberry Pi in the base

Level	What it shows	Status
0 · Digital design	CAD, calculations and FEA; logic flags 4 of 4 simulated spills	● Done
1 · 1:5 LIVE model	37 printed parts; the float rides its pole; real sensors at 1:1 in a dish	● Printed · assembly now
2 · Electronics on the model	Raspberry Pi reads, decides, acts: beacon, pump, SMS; wet test with 4 samples	● Before 19 Nov
3 · Tank test	Full-size float on a 2-inch pole: draft, 2 m travel, roller friction	● Planned
4 · Godavari pilot	30 days at Kopargaon; alerts checked against lab samples	● After permits

The LIVE model carries the real electronics:
it proves the measurement chain, not river-scale performance.

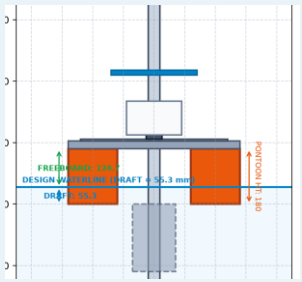
A10 · 66 parts, one mould, off-the-shelf electronics



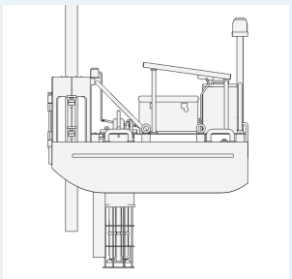


PE hulls	UV-stabilised, rotomoulded: one mould makes both hulls
Aluminium + HDPE	carriage: bolted aluminium angle; deck: HDPE marine board
Galvanised steel	pole and brackets: 2-inch GI pipe, galvanised plate (production)
Off the shelf	Raspberry Pi 3B+, sensor boards, SIM800L, 20 W solar, LiFePO4 battery

CAD caught our two biggest mistakes



v2 (old values): pole through the pod



R3: pole at the bow

Real mass from CAD: 25.4 → 48.5 kg, so the hulls grew (draft 55 → 117 mm)

Assembled in SOLIDWORKS; SOLIDWORKS Simulation, Pack and Go, ENOVIA Project Planner